

Badi Mirzai

Data Engineer

Stockholm, Sweden

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Profile

Data and software engineer with an MSc in Systems, Control & Robotics (KTH) and a BSc in Electrical Engineering.

Several years of experience building and maintaining data-driven applications, data pipelines, ELT processes, SQL databases, and distributed backend systems in production environments. Strong in Python, SQL, Oracle PL/SQL, Microsoft SQL Server, data modelling, API development, and cloud infrastructure. Experienced in translating business requirements into maintainable technical solutions and working with stakeholders in consulting and enterprise environments.

Skills

Languages: Svenska (modersmål), English

Programming Languages: Python, Go, SQL, TypeScript, C, C#

Data Engineering: Data pipelines, ELT/ETL, data transformation, data integration, data modelling, data processing, data quality

Data & Databases: Oracle PL/SQL, Microsoft SQL Server, PostgreSQL, Snowflake, Redis, dbt

Backend & APIs: REST, GraphQL, gRPC, OAuth 2.0, microservices, API development, event-driven systems

Infrastructure: Docker, Kubernetes, Helm, Terraform, Azure DevOps, CI/CD, Linux

Cloud: Azure, GCP (exposure)

Analytics & ML: Machine learning, data analysis, statistical modelling, JAX, NumPyro, MATLAB

Tools: Git, structured logging, OpenTelemetry, Postman, Cypress

Experience

Founder & Software Engineer · *Cyrux Systems*

Jan 2026 – Present

Stockholm, Sweden

- Built **Architon**, an open-source Go rule engine for deterministic hardware architecture verification; designed data ingestion pipelines for KiCad BOMs and netlists with structured, CI-ready output.
- Implemented validation and data transformation workflows for hardware design data, producing structured outputs and automated quality checks.
- Developed **Architon Studio**, a React and Go web platform for importing, processing, and visualising system architecture data.

Data & Applications Engineer · *BWC*

Jan 2023 – Nov 2025

Haifa, Israel

- Designed and maintained backend and data-driven systems serving **1,000+ staff across 40+ departments**, with high availability and strict data correctness requirements.
- Built Python automation and ETL reporting pipelines that reduced manual processing time by approximately **60%** across 15+ long-lived production applications.
- Worked with SQL databases including **Oracle PL/SQL, Microsoft SQL Server, and PostgreSQL**, supporting data transformation, integration, querying, and data modelling.
- Managed Azure DevOps CI/CD pipelines and Linux-based distributed services supporting production data flows and enterprise applications.

- Worked with business and technical stakeholders to translate requirements into maintainable data and application solutions.
- Owned incident resolution and root-cause analysis for mission-critical enterprise applications, including issues involving data processing and distributed systems.

IT Consultant • *Netlight Consulting*

Jun 2022 – Dec 2022

Stockholm, Sweden

- Designed and implemented modular **data pipelines** for internal analytics and decision support using Terraform and GitLab CI/CD.
- Translated ambiguous stakeholder requirements into maintainable, version-controlled, and documented technical solutions.
- Worked in a consulting environment with changing customer and project requirements.

Backend Developer (Consultant) • *AFA Försäkring*

Sep 2021 – Jun 2022

Stockholm, Sweden – Regulated insurance environment

- Led backend modernisation and designed Go microservices handling real-time data flows in a regulated, production-critical environment.
- Built distributed communication layers using GraphQL, Redis, OAuth 2.0, and database integrations with emphasis on correctness, performance, and secure data exchange.
- Owned automated testing across backend, authentication, and end-to-end flows; deployed and operated services on Kubernetes.
- Drove migration from a legacy monolith toward distributed, independently deployable backend services and APIs.

AI Researcher • *SMaRC – Swedish Maritime Robotics Centre (KTH)*

Jan 2021 – Jun 2021

Stockholm, Sweden

- Developed a simulation framework for autonomous underwater vehicles integrating physical dynamics, multi-DOF motion, and sensor uncertainty models.
- Trained Lagrangian Neural Networks for model identification and control using JAX and NumPyro; work published as a peer-reviewed thesis.

Software Developer • *Fotspar*

May 2020 – Jan 2021

Stockholm, Sweden

- Led development of a production mobile application; owned architecture decisions, technical reviews, and cross-functional stakeholder communication.

Education

KTH Royal Institute of Technology

Stockholm, Sweden

MSc in Computer Science – Systems, Control & Robotics

2019 – 2021

Focus: machine learning, control systems, optimisation, robotics

BSc in Electrical Engineering

2016 – 2019

Selected Projects

Architon - github.com/badimirzai/architon-cli

Open-source Go rule engine for hardware architecture verification. KiCad BOM and netlist ingestion, data transformation, deterministic validation, and CI-ready processing.

Applied Estimation - [GitHub](#)

Kalman filters, particle filters, and estimation algorithms in Python and MATLAB for robotics and navigation, with focus on data processing and modelling.

Publications

- Physics-Informed Deep Learning for System Identification of Autonomous Underwater Vehicles: A Lagrangian Neural Network Approach.
- Behaviour-Aware Motion Planning for Autonomous Vehicles Incorporating Human Driving Style.